

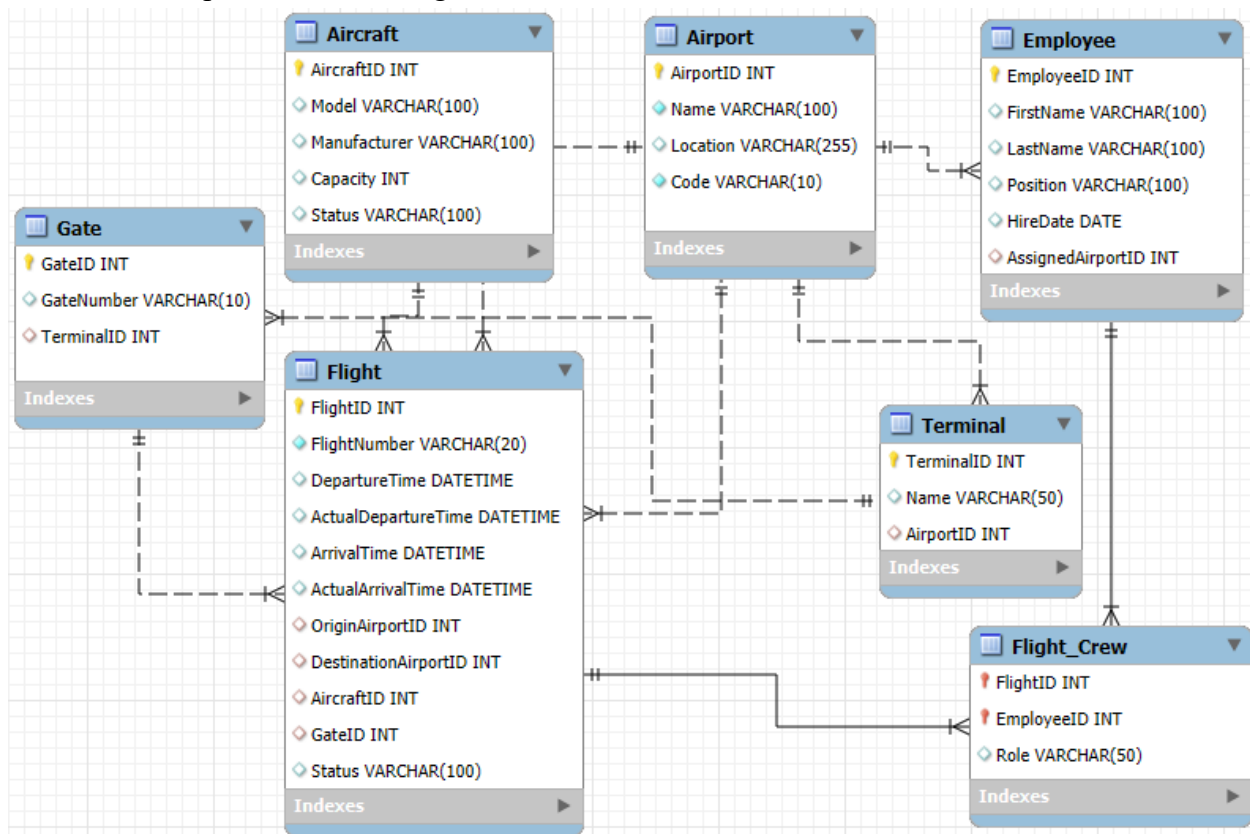
(Note: All materials are strictly forbidden during the examination.)

Section 1: (2.0 points) Theory:

- Describe main characteristics of object-relational databases.
- List main phases of creating and managing an object-relational database with object-oriented data models. Briefly explain each phase (1-2 sentences).

Section 2: (3.0 points) SQL

Given the Oracle Airport database design as below:



Note:

Terminal: Nhà ga, an airport can have many terminals.

Employee => Position: 'Pilot', 'Cabin Crew' or 'Ground Staff'

Flight => Status: 'Scheduled', 'Delayed', or 'Cancelled'

Aircraft => Status: 'Active', 'Busy', or 'Prepared'

Using SQL to complete the requests:

- Find the number of flights scheduled to depart from each airport today.
Note: Oracle uses TRUNC(DATE) to ignore the time part when comparing dates, using SYSDATE to get the current time.
- List all flights with their assigned pilot(s).
- Find the aircraft model with the highest number of flights assigned.

Section 3: (3.0 points) PL/SQL

Given the same database as section 2.

- a. Write a stored procedure `ADD_FLIGHT_WITH_DEFAULT_AIRCRAFT` that:
- Takes input: `FlightNumber`, `OriginAirportCode`, `DestinationAirportCode`, `DepartureTime`, `ArrivalTime`.
 - Finds the first available aircraft that is marked as 'Active'
 - Inserts a new record into the `Flight` table with the selected aircraft

If no active aircraft is found, raise an error.

- b. **Choose ONE of the following questions:**

Question c1:

Create a stored procedure `ASSIGN_PILOTS_TO_FLIGHT` that takes a `FlightID` as input and automatically assigns two available pilots who are:

- Assigned to the same airport as the flight's origin
- Not already assigned to another flight that overlaps in time

If fewer than 2 pilots are available, raise an exception.

Question c2:

Create a stored procedure `DISPLAY_FLIGHT_CREW` that shows information of flight crew of one given `FlightID`. Note that:

- Accepts `FlightID` as an input parameter of cursor
- Information show `FlightID`, `EmployeeID`, `FirstName`, `LastName`, and `Position`

Section 4: (2.0 points) Design

Show limitations of the `TeamMember` relation below. Then, propose improvements and briefly explain your ideas.

`TeamMember` (`TeamID`, `MemberID`, `Role`, `TeamName`, `SeasonID`, `MemberName`, `Birthday`, `Gender`, `Address`, `Phone`, `Email`)

Note that: One debate team (recognized by `TeamID`) has 3 members and register for a season. Each member of a team has a main role (including first speaker, second speaker, or third speaker).

----- End -----

(Note: The Proctor doesn't explain more)

Approved by Faculty

Lecturer